A Stylized Logit Demand Model with Internal Reference Pricing *

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Abstract

There is a growing literature applying the logit demand model to pharmaceutical markets. However, some pharmaceutical markets differ from typical empirical IO markets like breakfast cereals due to regulations such as copayment structures and internal reference prices. Therefore, it is important to assess the logit demand model's validity in such a complex context. After constructing a stylized logit demand model with an internal reference price, I use the intermediate value theorem to prove the existence of an equilibrium and identify conditions for equilibrium uniqueness in a 2-product market.

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1 Introduction

There are two main types of prescription medicines: originals (brand-name drugs) and generics. According to FDA (2024), a generic drug is "a medication created to be the same as an already marketed brand-name drug in dosage form, safety, strength, route of administration, quality, performance characteristics, and intended use". Because generics provide the same therapeutic effect as originals at a much lower price (Howland, 2009), regulators in many countries have introduced generic substitution to promote the use of generic drugs. The out-of-pocket expense is the sum of a copayment and the price difference between the chosen product's price and the reference price. The reference price can be determined by different ways. An external reference pricing system calculates the reference price through the prices of a basket of relevant countries. In an internal reference pricing regime, it is often the cheapest price among all substitutable products in the country. Consumers who agree to generic substitution only need to pay the copayment. If they oppose to generic substitution and prefer the prescribed product, they pay both the copayment and the price difference. The objective of generic substitution is to nudge patients towards affordable generic drugs and increase price competition among substitutable products.

As data shows price dispersion among substitutable product in pharmaceutical markets, there have been different approaches of explaining price differences. Some theoretical papers attribute price differences to horizontal differentiation in the spirit of Hotelling (1929). For example, Brekke et al. (2016) and Miraldo (2009) use Hotelling-style models to analyze the price equilibrium in the presence of a reference price. Although Hotelling-style models are useful in finding the price equilibrium in different settings, these models are not applicable to empirical studies. Many empirical papers in pharmaceutical markets (Dubois and Lasio, 2018; Kaiser et al., 2014; Janssen, 2023) assume existence and uniqueness of equilibrium in a logit demand model when estimating demand and conducting counterfactuals as the standard approach in empirical IO (Berry, 1994; Berry et al., 1995; Nevo, 2000, 2001). Equilibrium existence can be proven with the Schauder fixed-point theorem because the best-response mapping is continuous and compact. Uniqueness is also guaranteed if the reference price is exogenous or if the endogenous reference price cancels out in the market-share function.

When the market-share function includes an endogenous reference price, a unique equilibrium requires that all best-response functions intersect at a single point. Moreover, while firms receive the full wholesale price, consumers care only about out-of-pocket costs, which depend on the reference price. In addition, an outside good with normalized utility zero prevents us from canceling the reference price from the market-share function. The growing literature on pharmaceutical markets has not yet established whether an equilibrium exists

under these complications, and whether it is unique.

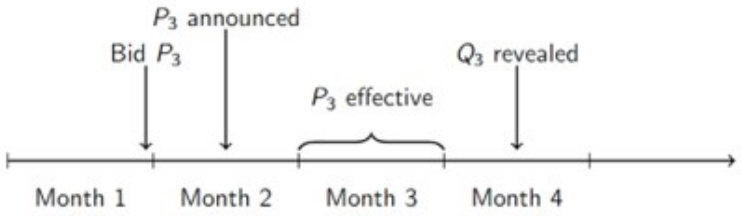
My contribution is to identify conditions for the existence and uniqueness of equilibrium in a logit model that incorporates an internal reference price and a copayment structure. Using the Intermediate Value Theorem, I prove equilibrium existence in a two-product market and show that the equilibrium is unique under additional restrictions: the two firms must be sufficiently different so that their best-response functions cross but do not overlap over an interval. I then extend the existence result to an n-product market with an aggregative-game approach.

The aim of this chapter is twofold. First, I enrich the theory of product differentiation with a stylized logit model that incorporates internal reference pricing. Second, by establishing equilibrium existence and uniqueness, the model provides a theoretical foundation for empirical studies of pharmaceutical markets with internal reference pricing.

The chapter proceeds as follows: Section 2 describes an example of internal reference pricing in Sweden's pharmaceutical market. Section 3 outlines my stylized logit demand model, which has a copayment structure and the outside good. Section 4 proves the existence of equilibrium and provides the conditions for equilibrium uniqueness. Section 5 extends the 2-product model to the n-product model. Section 6 concludes.

2 An example of Internal Reference Pricing in Sweden

Figure 1: Auction timeline



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To understand the need to include an internal reference price in the logit demand model, I first present an example of internal reference pricing in Sweden's pharmaceutical market.

Before October 1, 2009, pharmacies in Sweden were obligated to dispense the cheapest substitutable generic product they had in stock. In a pharmaceutical market reform effective since October 1, 2009, the process of generic substitution became centralized through a national auction. Every month, pharmaceutical firms submit the prices of their products two months in advance. Then, the Swedish Dental and Pharmaceutical Benefits Board selects the cheapest product for each substitution group. A substitution group includes products

with the same active ingredient, the same strength, and the same package size group. The timeline of the auction is depicted in Figure 1. Regardless of the auction outcomes, firms are required to commit to sell at the prices they have posted. The cheapest product, known as the 'Product of the month' (POTM), must be dispensed by all pharmacies unless it is objected to by prescribers or consumers. Due to a fixed scheme of markups applied to pharmaceutical products in Sweden, pharmacies do not have any incentives to sell particular products. More information about Swedish pharmaceutical market can be found in the 2017 report of TLV (2017).

The copayment rate of an individual depends on annual consumption. Consumers pay the full amount of pharmaceuticals up to SEK 1100 (≈ €98). Between SEK 1100 (≈ €98) and 2200 (≈ €196), the copayment rate is 50%, 25%, or 10%. The copayment rate is 0% once consumers reach an annual consumption of SEK 2200 (≈ €196).

The out-of-pocket expense is the sum of the copayment (copayment rate multiplied by the cheapest price/POTM price) and the price difference between the chosen product's posted price and the POTM price. Consumers who obtain the POTM only need to pay the copayment. If they oppose to generic substitution and prefer to purchase the prescribed product, they need to pay both the copayment and the price difference. If they wish to acquire a product that is neither the POTM nor the prescribed product, they need to pay the full amount. However, this last case rarely occurs in practice.

There are some caveats to the regulation that I do not model. If the prescribing physician opposes generic substitution due to medical reasons, consumers do not pay the price difference, even though they receive more expensive products. Pharmacies might oppose the substitution to POTM because it is not in stock. In that case, it also does not increase the out-of-pocket expenses for consumers. Additionally, pharmacies can sell the remainder of the last month's stock during the first two weeks of the new month (at last month's price).

3 Stylized logit demand model

I begin by building a stylized logit demand model with 2 products and the outside option. Section 5 extends the model to the n-product case.

3.1 Utility function

$$U_{ij} = \alpha - \beta \tilde{P}_j + \xi_j + \epsilon_{ij} \tag{1}$$

ξ_j is the unobserved quality (vertical differentiation). ξ_j is unobserved by econometricians but known to all firms. ϵ_{ij} is consumers' taste heterogeneity (horizontal differentiation). ϵ_{ij}

is unknown to both econometricians and firms. $\tilde{P}_j$ is the out-of-pocket expense.

The out-of-pocket expense is the sum of the co-payment and the price difference between the chosen product's price and the reference price (the cheapest price among all products). With a copayment rate ρ ($0 \leq \rho \leq 1$) and a reference price $P_r = min\{P_j\}$, a consumer pays $\rho P_r + (P_j - P_r)$ to buy product j.

3.2 Market share

I apply the logit demand model to obtain the market share functions. The reference price is the cheaper price $P_r = min\{P_1, P_2\}$. Therefore, I have 2 cases of product 1's demand when $P_2 = x$. The two market share functions are the same when $P_1 = x$.

If $\boldsymbol{P_1 \leq x}$

$$s_1 = \frac{exp(\alpha - \beta\rho P_1 + \xi_1)}{1 + exp(\alpha - \beta\rho P_1 + \xi_1) + exp(\alpha - \beta\rho P_1 - \beta(x - P_1) + \xi_2)} \tag{2}$$

If $\boldsymbol{P_1 \geq x}$

$$s_1 = \frac{exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1) + exp(\alpha - \beta\rho x + \xi_2)} \tag{3}$$

3.3 Profit-maximizing price

The profit maximization problems are as follows:

$$\max_{P_i}(P_i - c_i)s_i \tag{4}$$

A profit-maximizing price (if exists) must satisfy the first-order condition of the profit maximization problems. In the standard logit demand model, the unique profit-maximizing price is $P_i = c_i + \frac{1}{\beta(1-s_i)}$. Proposition 1 shows the profit-maximizing price function with the co-payment rate and the internal reference price in my model. Appendix A.1 solves the FOCs of the profit maximization problems. Appendix A.2 shows the quasiconcavity of the profit functions and guarantees that the FOC's solution is the profit-maximizing price.

Proposition 1 (Profit-maximizing price): *The profit-maximizing price of Product 1 when $P_2 = x$:*

If $\boldsymbol{P_1 \leq x}$

$$P_1 = f_L(x) = c_1 + \frac{1}{\beta\rho(1 - s_1) + \beta(1 - \rho)s_2} \tag{5}$$

If $P_1 \geq x$

$$P_1 = f_H(x) = c_1 + \frac{1}{\beta(1 - s_1)} \tag{6}$$

The two price functions are the same when $P_1 = x$.

Using the lambertW function as in Aravindakshan and Ratchford (2011), I rewrite the profit-maximizing prices of Product 1 in Proposition 2 (Proof in Appendix A.3).

Proposition 2 (Profit-maximizing price): *The profit-maximizing price of Product 1 when $P_2 = x$:*

If $P_1 \leq x$

$$exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta x + \beta f_L(x) + \xi_2) = exp(\alpha + \xi_1) \tag{7}$$

If $P_1 \geq x$

$$f_H(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}W(\frac{exp(\alpha + \xi_1 + \beta(1 - \rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) \tag{8}$$

The two price functions are the same when $P_1 = x$.

3.4 Best-response functions

Proposition 2 says that when Product 1 is the reference product, the profit-maximizing price of Product 1 is $f_L(x)$. However, the winning-profit-maximizing price $f_L(x)$ might be larger than x. In that case, Product 1 cannot have the optimal winning price $f_L(x)$ and win at the same time. To find the best-response of Product 1, I need to analyze Product 1's winning profit curve and losing profit curve. Proposition 3 outlines the best-response functions of Product 1 and Product 2.

Proposition 3 (Best-response functions):

For Product 1: *The best-response function is*

$$P_1(x) = f(x) = \begin{cases} f_H(x) & \text{if } x \leq H_1 \\ x & \text{if } H_1 \leq x \leq L_1 \\ f_L(x) & \text{if } L_1 \leq x \end{cases} \tag{9}$$

where $f_H(x)$ and $f_L(x)$ are from Proposition 2:

$$f_H(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}W(\frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) \tag{10}$$

$$exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta x + \beta f_L(x) + \xi_2) = exp(\alpha + \xi_1) \tag{11}$$

L_1 is the point where $f_L(x)$ crosses the 45-degree line $L_1 = f_L(L_1)$.
H_1 is the point where $f_H(x)$ crosses the 45-degree line $H_1 = f_H(H_1)$.
For Product 2: *The best-response function is*

$$P_2(x) = g(x) = \begin{cases} g_H(x) & \text{if } x \le H_2 \\ x & \text{if } H_2 \le x \le L_2 \\ f_L(x) & \text{if } L_2 \le x \end{cases} \tag{12}$$

where $g_H(x)$ and $g_L(x)$ are from Proposition 2:

$$g_H(x) = c_2 + \frac{1}{\beta} + \frac{1}{\beta}W(\frac{exp(\alpha + \xi_2 + \beta(1-\rho)x - \beta c_2 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_1)}) \tag{13}$$

$$exp(\beta\rho g_L(x))(\beta\rho g_L(x) - \beta\rho c_2 - 1) + (\beta g_L(x) - \beta c_2 - 1)exp(\alpha - \beta x + \beta g_L(x) + \xi_1) = exp(\alpha + \xi_2) \tag{14}$$

L_2 is the point when $g_L^{-1}(x)$ crosses the 45-degree line. $L_2 = g_L^{-1}(L_2)$.
H_2 is the point when $g_H^{-1}(x)$ crosses the 45-degree line $H_2 = g_H^{-1}(H_2)$.

Proof of Proposition 3. Best-response functions

Part 1: The shapes of the winning profit curve and the losing profit curve

I formulate the best-response function of Product 1 by drawing Product 1's winning profit curve and losing profit curve (Figure 3).

From Proposition 2,

$$(\beta f_H(x) - \beta c_1 - 1)exp(\beta f_H(x) - \beta c_1 - 1) = \frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} \tag{15}$$

$$\beta f_H(x) - \beta c_1 - 1 = \frac{exp(\alpha + \xi_1 + \beta(1-\rho)x) - \beta f_H(x)}{1 + exp(\alpha - \beta\rho x + \xi_2)} \tag{16}$$

$$(\beta f_H(x) - \beta c_1 - 1)(1 + exp(\alpha - \beta\rho x + \xi_2)) = exp(\alpha + \xi_1 + \beta(1-\rho)x) - \beta f_H(x) \tag{17}$$

$$exp(\beta\rho x)(\beta f_H(x) - \beta c_1 - 1) + (\beta f_H(x) - \beta c_1 - 1)exp(\alpha + \xi_2) = exp(\alpha - \beta(f_H(x) - x) + \xi_1) \tag{18}$$

If $f_H(x) > x$,

$$\begin{aligned} &exp(\beta\rho x)(\beta x - \beta c_1 - 1) + (\beta x - \beta c_1 - 1)exp(\alpha + \xi_2) \\ &\quad < exp(\beta\rho x)(\beta f_H(x) - \beta c_1 - 1) + (\beta f_H(x) - \beta c_1 - 1)exp(\alpha + \xi_2) \\ &\quad = exp(\alpha - \beta(f_H(x) - x) + \xi_1) < exp(\alpha + \xi_1) \end{aligned} \tag{19}$$

From Proposition 2,

$$\begin{aligned} &exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta x + \beta f_L(x) + \xi_2) \\ &\quad = exp(\alpha + \xi_1) \end{aligned} \tag{20}$$

If $f_L(x) < x$,

$$\begin{aligned} &exp(\beta\rho x)(\beta\rho x - \beta\rho c_1 - 1) + (\beta x - \beta c_1 - 1)exp(\alpha + \xi_2) \\ &> exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta x + \beta f_L(x) + \xi_2) \\ &\quad = exp(\alpha + \xi_1) \end{aligned} \tag{21}$$

If $f_L(x) < x \leq f_H(x)$ or $f_L(x) \leq x < f_H(x)$, I apply Equation 19 and Equation 21 to have:

$$\begin{aligned} &exp(\beta\rho x)(\beta\rho x - \beta\rho c_1 - 1) + (\beta x - \beta c_1 - 1)exp(\alpha + \xi_2) > \\ &\quad exp(\beta\rho x)(\beta x - \beta c_1 - 1) + (\beta x - \beta c_1 - 1)exp(\alpha + \xi_2) \end{aligned} \tag{22}$$

This is a contradiction because $\beta\rho x - \beta\rho c_1 \leq \beta x - \beta c_1$. Therefore, I can only have 3 cases:

Case 1: $f_H(x) \geq x$, $f_L(x) \geq x$ (Figure 2a)

The maximum points of the two profit curves lie on the right side of x. Therefore, the profit function is decreasing on the left of x and increasing on the right of x. It is optimal to price at $f_H(x)$.

Case 2: $f_H(x) \leq x \leq f_L(x)$ (Figure 2b)

The maximum point of the profit curve on the right side of x lies on the left side of x,

and vice versa. Therefore, the profit function is decreasing on both sides of x. It is optimal to price at x.

Case 3: $f_H(x) \leq x$, $f_L(x) \leq x$ (Figure 2c) The maximum points of the two profit curves lie on the left side of x. Therefore, the profit function is increasing on the left of x and decreasing on the right of x. It is optimal to price at $f_L(x)$.

Figure 2: Profit-maximizing price

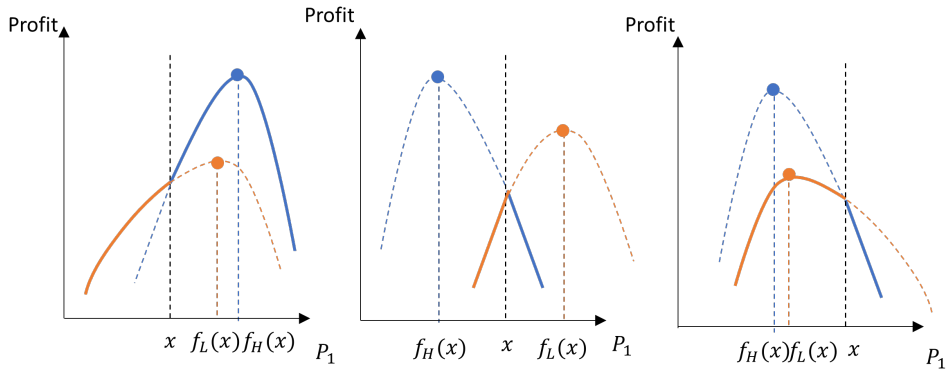


Figure 2a *Figure 2b* *Figure 2c*

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The best-response function of product 1 is given by:

$$P_1(x) = f(x) = \begin{cases} f_H(x) & \text{if } f_H(x) \geq x \text{ and } f_L(x) \geq x \\ x & \text{if } f_H(x) \leq x \leq f_L(x) \\ f_L(x) & \text{if } f_H(x) \leq x \text{ and } f_L(x) \leq x \end{cases} \tag{23}$$

where $f_H(x)$ and $f_L(x)$ are from Proposition 2.

Part 2: Rewriting the best-response functions

Because the conditions for the 3 cases of the best-response function which are prices endogenous to the model, I will rewrite the conditions using the intersections of the best-response curves and the 45-degree line.

Because $0 < \frac{df_H(x)}{dx}$ (Appendix A.4), the equation $f_H(x) - x = 0$ has a unique solution. $f_H(x)$ intersects the 45-degree line at most once. Similarly, $f_L(x)$ also intersects the 45-degree line at most once due to $0 < \frac{df_L(x)}{dx}$ (Appendix A.4).

Using Equation 18, $f_H(x)$ intersects the 45-degree line at point (H_1, H_1) if the following equations hold:

$$\exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) + (\beta H_1 - \beta c_1 - 1)\exp(\alpha + \xi_2) = \exp(\alpha + \xi_1) \tag{24}$$

Using Proposition 2, $f_L(x)$ intersects the 45-degree line at point (L_1, L_1) if:

$$\exp(\beta\rho L_1)(\beta\rho L_1 - \beta\rho c_1 - 1) + (\beta L_1 - \beta c_1 - 1)\exp(\alpha + \xi_2) = \exp(\alpha + \xi_1) \tag{25}$$

From the two equations above, I have:

$$\begin{aligned}\exp(\beta\rho L_1)(\beta\rho L_1 - \beta\rho c_1 - 1) + (\beta L_1 - \beta c_1 - 1)\exp(\alpha + \xi_2) = \\ \exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) + (\beta H_1 - \beta c_1 - 1)\exp(\alpha + \xi_2)\end{aligned} \tag{26}$$

$$\exp(\beta\rho L_1)(\beta\rho L_1 - \beta\rho c_1 - 1) + (\beta L_1 - \beta H_1)\exp(\alpha + \xi_2) - \exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) = 0 \tag{27}$$

If $L_1 < H_1$:

$$\begin{aligned}\exp(\beta\rho L_1)(\beta\rho L_1 - \beta\rho c_1 - 1) + (\beta L_1 - \beta H_1)\exp(\alpha + \xi_2) - \exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) \\ < \exp(\beta\rho H_1)(\beta\rho H_1 - \beta\rho c_1 - 1) + (\beta\rho H_1 - \beta\rho H_1)\exp(\alpha + \xi_2) \\ - \exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) < 0\end{aligned} \tag{28}$$

This is a contradiction with Equation 27. Therefore, $H_1 \leq L_1$.
Because $0 < \frac{df_L(x)}{dx}$ (Appendix A.4) and $f_L(L_1) = L_1$, I have $f_L(x) \geq x$ if $x \leq L_1$.
Because $0 < \frac{df_H(x)}{dx}$ (Appendix A.4) and $f_H(H_1) = H_1$, I have $f_H(x) \geq x$ if $x \leq H_1$.
Therefore, $f_H(x) \leq x$ and $f_L(x) \leq x$ if $x \geq L_1 \geq H_1$.
$f_H(x) \geq x$ and $f_L(x) \geq x$ if $x \leq H_1 \leq L_1$.
I can rewrite the best-response function of Product 1 as:

$$P_1(x) = f(x) = \begin{cases} f_H(x) & \text{if } x \leq H_1 \\ x & \text{if } H_1 \leq x \leq L_1 \\ f_L(x) & \text{if } x \geq L_1 \end{cases} \tag{29}$$

Similarly, I have the best-response function of Product 2:

$$P_2(x) = g(x) = \begin{cases} g_H(x) & \text{if } x \leq H_2 \\ x & \text{if } H_2 \leq x \leq L_2 \\ g_L(x) & \text{if } x \geq L_2 \end{cases} \tag{30}$$

QED.

It is intuitive that the best-response function is increasing in the opponent price (proof of Proposition 4 in Appendix A.4).

Proposition 4 (Increasing best-response functions): *The best-response function of each product is increasing in the opponent price.*

Because each best-response curve intersects the 45-degree line at most once in the intervals $[H_1, L_1]$ and $[H_2, L_2]$ respectively, I can find the equilibrium by identifying the intersection of the two best-response curves.

4 Equilibrium

In this section, I will first prove equilibrium existence, necessary and sufficient conditions for equilibrium uniqueness in the simplest case when both best-response curves cross the 45-degree line. I will also show the results still hold in the case of the best-response curves not crossing the 45-degree line.

4.1 Equilibrium existence

Proposition 5 (Equilibrium existence): *The two best-response curves have at least one intersection.*

Proof of Proposition 5. Equilibrium existence

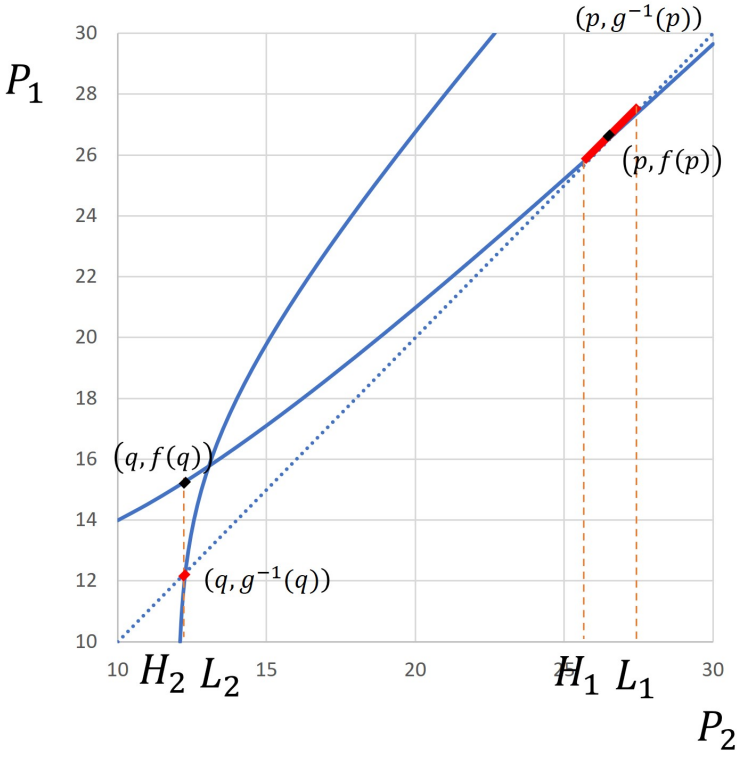
To have the two best-response functions in the same graph, I plot f(x) and the inverse function $g^{-1}(x)$. I will prove that $f(x)$ and $g^{-1}(x)$ have at least one intersection because $f(x)$ is an increasing convex function, $g^{-1}(x)$ is an increasing concave function, and both $f(x)$ and $g^{-1}(x)$ intersect the 45-degree line once.

If the intervals $[H_1, L_1]$ and $[H_2, L_2]$ overlap, $f(x)$ and $g^{-1}(x)$ will also overlap on the 45-degree line. This indicates the presence of multiple equilibria with equal prices.

If the intervals $[H_1, L_1]$ and $[H_2, L_2]$ do not overlap, without loss of generality, I assume that $L_2 < H_1$ (product 2's best-response curve $g^{-1}(x)$ crosses the 45-degree line first).

Let's denote point (p, p) belonging to the interval where $f(x)$ crosses the 45-degree line, and point (q, q) belonging to the interval where $g^{-1}(x)$ crosses the 45-degree line

Figure 3: Equilibrium existence



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Before crossing the 45-degree line, both best-response curves $f(x)$ and $g^{-1}(x)$ starts in the area where their price is higher than the opponent's price ($f(x)$ is above the 45-degree line, and $g^{-1}(x)$ is below the 45-degree line). Therefore, $f(q) > q$. Also, $q = g^{-1}(q)$. Hence, $f(q) > g^{-1}(q)$.

Since $g^{-1}(x)$ only crosses the 45-degree line once, $g^{-1}(x) \geq x$ for $x > q$. Therefore, $g^{-1}(p) \geq p$. Also, $f(p) = p$. Hence, $g^{-1}(p) \geq f(p)$.

I have $f(q) > g^{-1}(q)$ and $f(p) \leq g^{-1}(p)$ for $q < p$. Since both $f(.)$ and $g^{-1}(.)$ are continuous functions, they must intersect each other at least once in the interval (q, p) due to the Intermediate Value Theorem. QED.

The case of the best-response curves not crossing the 45-degree line

I denote $(x, g^{-1}(x))$ as the starting point of Product 2's best-response curve. I have $g^{-1}(x) = c_1$ because the starting point is Product 2's best response to $P_1 = c_1$.

The best response always guarantees a positive profit for Product 2: $x > c_2$. Because the best-response function of Product 1 is weakly increasing in P_2, $f(x) \geq f(c_2)$. Also, the best response always guarantees a positive profit for Product 1: $f(c_2) > c_1$. Therefore, $f(x) > c_1$ or $f(x) > g^{-1}(x)$.

In order to have $f(x) > g^{-1}(x)$, at least one best-response function has to cross the 45-degree line. If both best-response functions do not cross the 45-degree line, $f(a) < g^{-1}(a)$ at

all values of a because $f(a)$ is always below the 45-degree line and $g^{-1}(a)$ is always above. the 45-degree line.

Similar to the first case of both best-response curves crossing the 45-degree line, $f(H_1) \leq g^{-1}(H_1)$. I have $f(x) > g^{-1}(x)$ and $f(H_1) \leq g^{-1}(H_1)$ for $x < H_1$. Since both $f(.)$ and $g^{-1}(.)$ are continuous functions, they must intersect each other at least once in the interval (x, H_1) due to the Intermediate Value Theorem.

4.2 Equilibrium uniqueness conditions

Proposition 6 (Equilibrium uniqueness conditions): *The equilibrium is unique if both best-response curves cross the 45-degree line and the intersection points satisfies either $L_2 < H_1$ or $L_1 < H_2$*

H_1, L_1, H_2, L_2 are solutions to the following system of equations:

$$\begin{cases} exp(\beta\rho H_1)(\beta H_1 - \beta c_1 - 1) + (\beta H_1 - \beta c_1 - 1)exp(\alpha + \xi_2) = exp(\alpha + \xi_1) \\ exp(\beta\rho L_1)(\beta\rho L_1 - \beta\rho c_1 - 1) + (\beta L_1 - \beta c_1 - 1)exp(\alpha + \xi_2) = exp(\alpha + \xi_1) \\ exp(\beta\rho H_2)(\beta H_2 - \beta c_2 - 1) + (\beta H_2 - \beta c_2 - 1)exp(\alpha + \xi_1) = exp(\alpha + \xi_2) \\ exp(\beta\rho L_2)(\beta\rho L_2 - \beta\rho c_2 - 1) + (\beta L_2 - \beta c_2 - 1)exp(\alpha + \xi_1) = exp(\alpha + \xi_2) \end{cases} \tag{31}$$

Proof of Proposition 6. Equilibrium uniqueness

In the previous subsection, I have proven that $f(x)$ and $g^{-1}(x)$ intersect at least once. If and only if the intervals $[H_1, L_1]$ and $[H_2, L_2]$ overlap, $f(x)$ and $g^{-1}(x)$ overlap on the 45-degree line, which indicates the presence of multiple equilibria with equal prices.

If and only if the intervals $[H_1, L_1]$ and $[H_2, L_2]$ do not overlap, the two best-response curves intersect at a point that is not on the 45-degree line. After this intersection, the slope of $f(x)$ is greater than the slope of $g^{-1}(x)$ because except for those points on the 45-degree line, $f(x)x < 1$ and $g^{-1}(x)x > 1$ (proof can be found in Appendix A.4). Therefore, the two best-response curves will not intersect again, and the equilibrium is unique.

If one of the two best-response curves does not cross the 45-degree line, the equilibrium is unique because the two best-response curves do not overlap on the 45-degree line.

In summary, I have multiple equilibria with equal prices if and only $f(x)$ overlaps with $g^{-1}(x)$ on the 45-degree line (if and only if $L_2 \geq H_1$ and $L_1 \geq H_2$). I have a unique equilibrium with different prices if and only if $L_2 < H_1$ or $L_1 < H_2$. QED.

5 The n-product model

I will prove equilibrium existence in the n-product model by having an aggregate term of all losing products (not the cheapest product).

Proposition 7 (n-product equilibrium existence):

There exists an aggregate term Σ that is consistent with the equilibrium cheapest price $P_w(\Sigma)$ and all other prices $P_j(\Sigma)$ such that:

$$\Sigma = \sum_{j \neq w} exp(\alpha - \beta\rho P_w(\Sigma) - \beta(P_j(\Sigma) - P_w(\Sigma)) + \xi_j \tag{32}$$

Proof of Proposition 7. n-product equilibrium existence

In part 1, I will show that it is possible to identify the cheapest product for each Σ based on the best response function of each product. In part 2 and 3, I will show that for each Σ, I can map it to the corresponding prices for all products $P_w(\Sigma)$ and $P_j(\Sigma)$. In part 4, I will show that there exists Σ that is consistent with all prices I found in equilibrium $P_w(\Sigma)$ and $P_j(\Sigma)$.

Part 1: existence of the cheapest product

I fix the aggregate term for all products except for the winning one:

$$\Sigma = \sum_{j \neq w} exp(\alpha - \beta\rho P_w - \beta(P_j - P_w) + \xi_j \tag{33}$$

To apply the 2-product model, think of the case when each product faces an imaginary aggregate product representing all other products, the lowest price of the aggregate product changes, but the aggregate term stays constant.

The price of the aggregate product (the lowest among other products) is x. The best response function when product j has a higher price than the aggregate product:

$$P_j = c_j + \frac{1}{\beta(1 - s_j)} \tag{34}$$

$$P_j = c_j + \frac{1}{\beta(1 - \frac{exp(\alpha - \beta\rho x - \beta(P_j - x) + \xi_j)}{1 + exp(\alpha - \beta\rho x - \beta(P_j - x)) + \Sigma}} \tag{35}$$

I find the first intersection point of Product j's best-response function and the 45-degree line:

$$H_j = c_j + \frac{1}{\beta(1 - \frac{exp(\alpha - \beta\rho H_j + \xi_j))}{1 + exp(\alpha - \beta\rho H_j + \xi_j) + \Sigma}} \tag{36}$$

$$H_j = c_j + \frac{1 + \Sigma + exp(\alpha - \beta\rho H_j + \xi_j)}{\beta(1+\Sigma)} \tag{37}$$

$$H_j = c_j + \frac{1}{\beta} + \frac{exp(\alpha - \beta\rho H_j + \xi_j)}{\beta(1+\Sigma)} \tag{38}$$

$$1 + \Sigma = \frac{exp(\alpha - \beta\rho H_j + \xi_j)}{\beta(H_j - c_j) - 1} \tag{39}$$

Because function $\frac{exp(\alpha-\beta\rho H_j+\xi_j)}{\beta(H_j-c_j)-1}$ is increasing in H_j, there exists a unique H_j for each Σ. The product with the lowest H_j is the product whose the best response function crossing the 45-degree line first. Therefore, for each Σ, I can identify the product with the cheapest price in equilibrium.

Part 2: Mapping the aggregate term to the cheapest price in equilibrium

I call the product with the cheapest price in equilibrium with the given Σ Product w.

$$P_w = c_w + \frac{1}{\beta\rho(1 - s_w) + \beta(1-\rho)\sum_{j \neq w} s_j} \tag{40}$$

$$P_w = c_w + \frac{1}{\beta\rho\frac{1+\Sigma}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma} + \beta(1-\rho)\frac{\Sigma}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma}} \tag{41}$$

Dividing both sides by P_w

$$\Omega_w(P_w) = 1 = \frac{c_w}{P_w} + \frac{1}{\beta\rho P_w\frac{1}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma} + \beta P_w\frac{\Sigma}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma}} \tag{42}$$

Given Σ, when $P_w \to \infty$, $\Omega_w(P_w) \to 0$.

Given Σ, when $P_w \to c_w$,

$\Omega_w(P_w) \to 1 + \frac{1}{\beta\rho c_w\frac{1}{1+exp(\alpha-\beta\rho c_w+\xi_w)+\Sigma} + \beta c_w\frac{\Sigma}{1+exp(\alpha-\beta\rho c_w+\xi_w)+\Sigma}} > 1$

Therefore, for all Σ, there exists P_w such that $\Omega_w(P_w) = 1$.

Part 3: Mapping the aggregate term to other prices in equilibirum

$$P_j = c_j + \frac{1}{\beta(1 - s_j)} \tag{43}$$

$$P_j = c_j + \frac{1}{\beta(1 - \frac{exp(\alpha-\beta\rho P_w-\beta(P_j-P_w)+\xi_j)}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma})} \tag{44}$$

Dividing both sides by P_j

$$\Omega_j(P_j) = 1 = \frac{c_j}{P_j} + \frac{1}{\beta P_j(1 - \frac{exp(\alpha-\beta\rho P_w-\beta(P_j-P_w)+\xi_j)}{1+exp(\alpha-\beta\rho P_w+\xi_w)+\Sigma})} \tag{45}$$

Given Σ, when $P_j \to \infty$, $\Omega_j(P_j) \to 0$.

Given Σ, when $P_j \to c_j$,

$\Omega_j(P_j) \to 1 + \frac{1}{\beta c_j(1 - \frac{exp(\alpha - \beta\rho P_w - \beta(c_j - P_w) + \xi_j)}{1 + exp(\alpha - \beta\rho P_w + \xi_w) + \Sigma})} > 1$

Therefore, for all Σ, there exists P_j such that $\Omega_j(P_j) = 1$.

Part 4: Existence of the aggregate term

The aggregate term:

$$\Sigma = \sum_{j \neq w} exp(\alpha - \beta\rho P_w - \beta(P_j - P_w) + \xi_j \tag{46}$$

Dividing both sides by Σ

$$\Omega_\Sigma(P_w, P_j) = 1 = \frac{\sum_{j \neq w} exp(\alpha - \beta\rho P_w - \beta(P_j - P_w) + \xi_j}{\Sigma} \tag{47}$$

when $\Sigma \to \infty$, $P_w, P_j \to$ finite prices:

$$P_w \to c_w + \frac{1}{\beta} \tag{48}$$

$$P_j \to c_j + \frac{1}{\beta} \tag{49}$$

Therefore, $\Omega_\Sigma(P_w, P_j) \to 0$

when $\Sigma \to 0$, $P_w, P_j \to$ finite prices:

$$P_w \to c_w + \frac{1}{\beta\rho \frac{1}{1 + exp(\alpha - \beta\rho P_w + \xi_w)}} \tag{50}$$

I show that the limit is finite by defining a function $\varphi(P_w)$ as:

$$\varphi(P_w) = \beta\rho P_w - \beta\rho c_w - exp(\alpha - \beta\rho P_w + \xi_w) \tag{51}$$

The limit is finite if I can find a finite P_w such that $\varphi(P_w) = 1$.

$\varphi(P_w)$ is increasing in P_w because

$$\frac{d\varphi(P_w)}{dP_w} = \beta\rho + \beta\rho exp(\alpha - \beta\rho P_w + \xi_w) > 0 \tag{52}$$

When $P_w \to \infty$, $\varphi(P_w) \to \infty$. When $P_w \to c_w$, $\varphi(P_w) \to -exp(\alpha - \beta\rho c_w + \xi_w)$. Therefore, there always exists a finite P_w such that $\varphi(P_w) = 1$. Hence, there exists a finite P_w such that $P_w \to$ finite $c_w + \frac{1}{\beta\rho \frac{1}{1 + exp(\alpha - \beta\rho P_w + \xi_w)}}$ when $\Sigma \to 0$.

Because P_w is finite, all P_j are also finite.

$$P_j \to c_j + \frac{1}{\beta(1 - \frac{exp(\alpha - \beta\rho P_w - \beta(P_j - P_w) + \xi_j)}{1 + exp(\alpha - \beta\rho P_w + \xi_w)})} \tag{53}$$

Therefore, $\Omega_\Sigma(P_w, P_j) \to \infty$ when $\Sigma \to 0$.

Because $\Omega_\Sigma(P_w, P_j) \to 0$ when $\Sigma \to \infty$, there exists Σ such that $\Omega_\Sigma(P_w, P_j) = 1$. Then, for all Σ, I can find the corresponding P_w and all P_j. QED.

6 Conclusion

Using the intermediate value theorem, I have shown that there always exists an equilibrium in my stylized 2-product logit demand model with an internal reference price, and that the equilibrium is unique under additional conditions. I hope that my model can serve as a theoretical framework for the application of logit demand model in empirical studies on pharmaceutical markets with internal reference pricing.

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A Appendices

A.1 Profit-maximizing problems

Proof of Proposition 1. Profit-maximizing price

With $P_2 = x$, I solve for the profit-maximizing problems of Product 1 in 2 cases when $P_1 \leq x$ and $P_1 \geq x$.

Case 1: $P_1 \leq x$

$$\max_{P_1}(P_1 - c_1) * \frac{exp(\alpha - \beta\rho P_1 + \xi_1)}{1 + exp(\alpha - \beta\rho P_1 + \xi_1) + exp(\alpha - \beta\rho P_1 - \beta(x - P_1) + \xi_2)} \tag{54}$$

FOC:

$$s_1 + (P_1 - c_1) * (-\beta\rho)s_1 - (P_1 - c_1) * (-\beta\rho)s_1^2 - (P_1 - c_1) * (-\beta\rho + \beta)s_1 s_2 = 0 \tag{55}$$

$$P_1 = f_L(x) = c_1 + \frac{1}{\beta\rho(1 - s_1) + \beta(1 - \rho)s_2} \tag{56}$$

Case 2: $P_1 \geq x$

$$\max_{P_1}(P_1 - c_1) * \frac{exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2) + exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1)} \tag{57}$$

FOC:

$$s_1 + (P_1 - c_1) * (-\beta)s_1 - (P_1 - c_1) * (-\beta)s_1^2 = 0 \tag{58}$$

$$P_1 = f_H(x) = c_1 + \frac{1}{\beta(1 - s_1)} \tag{59}$$

QED.

A.2 Quasiconcavity of the profit functions

With $P_2 = x$, I show the quasiconcavity of the profit function of Product 1 in 2 cases when $P_1 \leq x$ and $P_1 \geq x$.

Case 1: $P_1 \leq x$

I need to prove

$$\begin{cases} \frac{\partial \Pi_1(P_1,x)}{\partial P_1} = 0 & \text{when } P_1 = f_L(x) \\ \frac{\partial \Pi_1(P_1,x)}{\partial P_1} > 0 & \text{when } P_1 < f_L(x) \\ \frac{\partial \Pi_1(P_1,x)}{\partial P_1} < 0 & \text{when } P_1 > f_L(x) \end{cases} \tag{60}$$

The profit function of Product 1 is:

$$\Pi_1(P_1, x) = (P_1 - c_1)\frac{exp(\alpha - \beta\rho P_1 + \xi_1)}{1 + exp(\alpha - \beta\rho P_1 + \xi_1) + exp(\alpha - \beta\rho P_1 - \beta(x - P_1) + \xi_2)} \tag{61}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} = s_1 + (P_1 - c_1)[(-\beta\rho)s_1 - (-\beta\rho)s_1^2 - (-\beta\rho + \beta)s_1 s_2] \tag{62}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} = s_1[1 + (P_1 - c_1)[-\beta\rho(1 - s_1) - \beta(1 - \rho)s_2]] \tag{63}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} > 0 \text{ if } 1 + (P_1 - c_1)[-\beta\rho(1 - s_1) - \beta(1 - \rho)s_2] < 0 \tag{64}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} > 0 \text{ if } P_1 < c_1 + \frac{1}{\beta\rho(1 - s_1) + \beta(1 - \rho)s_2} \tag{65}$$

When $P_1 < f_L(x)$, I have

$$s_1(P_1, x) > s_1^*(f_L(x), x) \text{ and } s_2(P_1, x) < s_2^*(f_L(x), x) \tag{66}$$

$$c_1 + \frac{1}{\beta\rho(1 - s_1) + \beta(1 - \rho)s_2} > c_1 + \frac{1}{\beta\rho(1 - s_1^*) + \beta(1 - \rho)s_2^*} = f_L(x) \tag{67}$$

$$c_1 + \frac{1}{\beta\rho(1 - s_1) + \beta(1 - \rho)s_2} > P_1 \tag{68}$$

Therefore, for any $P_1 < f_L(x)$, the first derivative $\frac{\partial \Pi_1(P_1,x)}{\partial P_1} > 0$.

Similarly, I can show $\frac{\partial \Pi_1(P_1,x)}{\partial P_1} < 0$ when $P_1 > f_L(x)$.

Case 2: $P_1 \geq x$

I need to prove

$$\begin{cases} \frac{\partial \Pi_1(P_1,x)}{\partial P_1} = 0 & \text{when } P_1 = f_H(x) \\ \frac{\partial \Pi_1(P_1,x)}{\partial P_1} > 0 & \text{when } P_1 < f_H(x) \\ \frac{\partial \Pi_1(P_1,x)}{\partial P_1} < 0 & \text{when } P_1 > f_H(x) \end{cases} \tag{69}$$

The profit function of Product 1 is:

$$\Pi_1(P_1, x) = (P_1 - c_1)\frac{exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x - \beta(P_1 - x) + \xi_1) + exp(\alpha - \beta\rho x + \xi_2)} \tag{70}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} = s_1 + (P_1 - c_1)[(-\beta)s_1 - (-\beta)s_1^2] \tag{71}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} = s_1[1 + (P_1 - c_1)[-\beta(1 - s_1)]] \tag{72}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} > 0 \text{ if } 1 + (P_1 - c_1)[-\beta(1 - s_1) < 0] \tag{73}$$

$$\frac{\partial \Pi_1(P_1, x)}{\partial P_1} > 0 \text{ if } P_1 < c_1 + \frac{1}{\beta(1 - s_1)} \tag{74}$$

When $P_1 < f_H(x)$, I have

$$s_1(P_1, x) > s_1^*(f_H(x), x) \tag{75}$$

$$c_1 + \frac{1}{\beta(1 - s_1)} > c_1 + \frac{1}{\beta(1 - s_1^*)} = f_H(x) \tag{76}$$

$$c_1 + \frac{1}{\beta(1 - s_1)} > P_1 \tag{77}$$

Therefore, for any $P_1 < f_H(x)$, the first derivative $\frac{\partial \Pi_1(P_1, x)}{\partial P_1} > 0$.

Similarly, I can show $\frac{\partial \Pi_1(P_1, x)}{\partial P_1} < 0$ when $P_1 > f_H(x)$. QED.

A.3 Profit-maximizing prices with the LambertW function

Proof of Proposition 2. Closed-form profit-maximizing price

With $P_2 = x$, I solve for the profit-maximizing prices of Product 1 in 2 cases when $P_1 \leq x$ and $P_1 \geq x$.

Case 1: $P_1 \leq x$

$$f_L(x) = c_1 + \frac{1}{\beta\rho(1-s_1) + \beta(1-\rho)s_2} = c_1 + \frac{1}{\beta s_2 + \beta\rho s_0} \tag{78}$$

$$f_L(x) = c_1 + \frac{1}{\beta \frac{\rho + exp(\alpha - \beta\rho f_L(x) - \beta(x - f_L(x)) + \xi_2)}{1 + exp(\alpha - \beta\rho f_L(x) + \xi_1) + exp(\alpha - \beta\rho f_L(x) - \beta(x - f_L(x)) + \xi_2)}} \tag{79}$$

$$f_L(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}\frac{1 - \rho + exp(\alpha - \beta\rho f_L(x) + \xi_1)}{\rho + exp(\alpha - \beta\rho f_L(x) - \beta(x - f_L(x)) + \xi_2)} \tag{80}$$

$$\begin{aligned}\rho(\beta f_L(x) - \beta c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta\rho f_L(x) - \beta(x - f_L(x)) + \xi_2) \\ = 1 - \rho + exp(\alpha - \beta\rho f_L(x) + \xi_1)\end{aligned} \tag{81}$$

$$\begin{aligned}exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)exp(\alpha - \beta x + \beta f_L(x)) + \xi_2) \\ = exp(\alpha + \xi_1)\end{aligned} \tag{82}$$

Case 2: $P_1 \geq x$

$$f_H(x) = c_1 + \frac{1}{\beta(1-s_1)} = c_1 + \frac{1}{\beta(s_2 + s_0)} \tag{83}$$

$$f_H(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}\frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} \tag{84}$$

Multiplying both sides by β and adding $\alpha + \xi_1 + \beta(1-\rho)x$

$$\begin{aligned}\beta f_H(x) + \alpha + \xi_1 + \beta(1-\rho)x \\ = \beta c_1 + 1 + \frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} + \alpha + \xi_1 + \beta(1-\rho)x\end{aligned} \tag{85}$$

$$\begin{aligned}\frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} + (\alpha - \beta f_H(x) + \beta(1-\rho)x + \xi_1) \\ = \alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1\end{aligned} \tag{86}$$

Taking exponential function of both sides

$$exp(\frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) * exp(\alpha - \beta f_H(x) + \beta(1-\rho)x + \xi_1)$$
$$= exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1) \quad (87)$$

Dividing both sides by $1 + exp(\alpha - \beta\rho x + \xi_2)$

$$exp(\frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} * \frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}$$
$$= \frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} \quad (88)$$

The left-hand side has the form $e^x * x = z$. I can use Lambert W function $W(z) = x$:

$$\frac{exp(\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1)}{1 + exp(\alpha - \beta\rho x + \xi_2)} = W(\frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) \quad (89)$$

I take natural log of both sides. The natural log of a Lambert W function is: $\ln W(z) = \ln z - W(z)$.

$$\alpha - \beta\rho x - \beta(f_H(x) - x) + \xi_1$$
$$= (\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1) - W(\frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) \quad (90)$$

$$f_H(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}W(\frac{exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + exp(\alpha - \beta\rho x + \xi_2)}) \quad (91)$$

QED.

A.4 Derivatives of best-response functions

Proof of Proposition 4. Derivatives of best-response functions

I prove that the first derivatives of all best-response functions are positive and smaller than 1.

Product 1

From Proposition 2, for $f_L(x)$:

$$\exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1 - 1) + (\beta f_L(x) - \beta c_1 - 1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2) = \exp(\alpha + \xi_1) \quad (92)$$

If $\beta f_L(x) - \beta c_1 - 1 \leq 0$, then $\beta\rho f_L(x) - \beta\rho c_1 - 1 \leq 0$ because $\beta\rho f_L(x) - \beta\rho c_1 \leq \beta f_L(x) - \beta c_1$. However, if $\beta f_L(x) - \beta c_1 - 1 \leq 0$ and $\beta\rho f_L(x) - \beta\rho c_1 - 1 \leq 0$, then the left-hand side is nonpositive while the right-hand side is positive. Therefore, $\beta f_L(x) - \beta c_1 - 1 > 0$

Differentiating both sides of Equation 92 with respect to x, I have:

$$\begin{aligned}\exp(\beta\rho f_L(x))\beta\rho\frac{df_L(x)}{dx}(\beta\rho f_L(x) - \beta\rho c_1 - 1) + \exp(\beta\rho f_L(x))\beta\rho\frac{df_L(x)}{dx} \\ + \beta\frac{df_L(x)}{dx}\exp(\alpha - \beta x + \beta f_L(x) + \xi_2) \\ + (\beta f_L(x) - \beta c_1 - 1)\beta\frac{df_L(x)}{dx}\exp(\alpha - \beta x + \beta f_L(x) + \xi_2) \\ + (\beta f_L(x) - \beta c_1 - 1)(-\beta)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2) = 0 \quad (93)\end{aligned}$$

$$\frac{df_L(x)}{dx}[\beta\rho\exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1) + \beta(\beta f_L(x) - \beta c_1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2)] = \beta(\beta f_L(x) - \beta c_1 - 1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2) \quad (94)$$

$\frac{df_L(x)}{dx} > 0$ because $\beta f_L(x) - \beta c_1 - 1 > 0$, $\beta f_L(x) - \beta c_1 > 0$, and $\beta\rho f_L(x) - \beta\rho c_1 > 0$.

$$\frac{df_L(x)}{dx} = \frac{(\beta f_L(x) - \beta c_1 - 1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2)}{\rho\exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1) + (\beta f_L(x) - \beta c_1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2)} \quad (95)$$

$$\frac{df_L(x)}{dx} - 1 = \frac{-\rho\exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1) - \exp(\alpha - \beta x + \beta f_L(x) + \xi_2)}{\rho\exp(\beta\rho f_L(x))(\beta\rho f_L(x) - \beta\rho c_1) + (\beta f_L(x) - \beta c_1)\exp(\alpha - \beta x + \beta f_L(x) + \xi_2)} < 0 \quad (96)$$

Therefore:

$$0 < \frac{df_L(x)}{dx} < 1 \quad (97)$$

From Proposition 2, for $f_H(x)$:

$$f_H(x) = c_1 + \frac{1}{\beta} + \frac{1}{\beta}W\left(\frac{\exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + \exp(\alpha - \beta\rho x + \xi_2)}\right) \quad (98)$$

$$\text{Let } A = \frac{\exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)}{1 + \exp(\alpha - \beta\rho x + \xi_2)} \quad (99)$$

For the lambert W function $W(z)$, its derivative is

$$W'(z) = \frac{W(z)}{z(1 + W(z))} \quad (100)$$

For the best-response function of product 1 $f_H(x)$:

$$\begin{aligned}\frac{df_H(x)}{dx} &= \frac{1}{\beta}\frac{W(A)}{A(1+W(A))} \\ &\left[\frac{\beta(1-\rho)\exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)[1 + \exp(\alpha - \beta\rho x + \xi_2)]}{[1 + \exp(\alpha - \beta\rho x + \xi_2)]^2}\right. \\ &\left. - \frac{\exp(\alpha + \xi_1 + \beta(1-\rho)x - \beta c_1 - 1)(-\beta\rho)\exp(\alpha - \beta\rho x + \xi_2)}{[1 + \exp(\alpha - \beta\rho x + \xi_2)]^2}\right] \\ &= \frac{1}{\beta}\frac{W(A)}{A(1+W(A))}\left[\beta(1-\rho)A + \beta\rho A\frac{\exp(\alpha - \beta\rho x + \xi_2)}{1 + \exp(\alpha - \beta\rho x + \xi_2)}\right] \\ \frac{df_H(x)}{dx} &= \frac{W(A)}{1+W(A)}\left[1 - \rho + \rho\frac{\exp(\alpha - \beta\rho x + \xi_2)}{1 + \exp(\alpha - \beta\rho x + \xi_2)}\right] \quad (101)\end{aligned}$$

$\frac{df_H(x)}{dx} > 0$ because $\rho \leq 1$

Also, $\frac{df_H(x)}{dx} < 1$ because $\frac{W(A)}{1+W(A)} < 1$ and $1 - \rho + \rho\frac{\exp(\alpha-\beta\rho x+\xi_2)}{1+\exp(\alpha-\beta\rho x+\xi_2)} < 1$

Therefore: $0 < \frac{df_H(x)}{dx} < 1$

Product 2

Similarly, I can show that:

$$0 < \frac{dg_L(x)}{dx} < 1, 0 < \frac{dg_g(x)}{dx} < 1 \tag{102}$$

and

$$\frac{dg_L^{-1}(x)}{dx} = \frac{1}{\frac{dg_L(x)}{dx}} > 1, \frac{dg_H^{-1}(x)}{dx} = \frac{1}{\frac{dg_H(x)}{dx}} > 1 \tag{103}$$

QED.